Endpoint multilinear restricted weak type extrapolation theorem

Kangwei Li

Zhejiang Normal University

Hunan Normal University
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About the topic

In harmonic analysis, most of the problems are reduced to proving the boundedness of some operator:

$$\|Tf\|_{L^p} \leq C\|f\|_{L^p}, \qquad 1 \leq p_0 < p < p_1 \leq \infty.$$

Usually for some 'p' (for instance $p = 2$) we might have more tools. For this reason, we want to further reduce the problem to studying some particular L^p estimate. This tool is called 'extrapolation theorem'.

Rubio de Francia extrapolation theorem

Theorem (Rubio de Francia, 1984)

Suppose that T is a bounded operator on $L^{p_0}(w)$ for every $w \in A_{p_0}$, where $1 \leq p_0 < \infty$. Then T is bounded on $L^p(w)$ for all $w \in A_p$, $1 < p < \infty$.

Recall that we say $w \in A_p$ if

$$[w]_{A_p} := \frac{1}{|Q|}\int_Q w(x)\mathrm{d}x \left(\frac{1}{|Q|}\int_Q w(x)^{-\frac{1}{p-1}}\mathrm{d}x\right)^{p-1} < \infty$$

for $p > 1$ and

$$[w]_{A_1} := \operatorname{ess\,sup} \frac{Mw}{w} < \infty$$

for $p = 1$.

Antonio Cordoba:

There are no L^p spaces, only weighted L^2.

Remarks

Several remarks are in order:

- (Cruz-Uribe and Pérez, 2000)The operator T does not play any role in the proof, so that one can reformulate it as

$$\|g\|_{L^{p_0}(w)} \leq C(p_0, [w]_{A_{p_0}})\|f\|_{L^{p_0}(w)}, \quad \forall w \in A_{p_0}$$

  implies that for all $1 < p < \infty$ and all $w \in A_p$,

$$\|g\|_{L^{p}(w)} \leq C(p, [w]_{A_{p}})\|f\|_{L^{p}(w)}.$$

- If we assume that T is sublinear, then $T : L^{p_0}(w) \to L^{p_0,\infty}(w)$ for all $w \in A_{p_0}$ and some $1 \leq p_0 < \infty$ implies that $T : L^p(w) \to L^p(w)$ for all $1 < p < \infty$ and all $w \in A_p$.

Remarks

- (Rubio de Francia, 1984) Rescaling: If $\|g\|_{L^{p_0}(w)} \leq C(p_0, [w]_{A_{p_0/r}})\|f\|_{L^{p_0}(w)}$ holds for some $p_0 \geq r$ and all $w \in A_{p_0/r}$, then we have for all $r < p < \infty$ and all $w \in A_{p/r}$

$$\|g\|_{L^p(w)} \leq C(p, [w]_{A_{p/r}})\|f\|_{L^p(w)}.$$

- (Auscher and Martell, 2007) Limited range: If $\|g\|_{L^q(w)} \leq C\|f\|_{L^q(w)}$ holds for some $q \in [p_0, q_0]$ and all $w \in A_{q/p_0} \cap RH_{(\frac{q_0}{q})'}$, then we have for all $p \in (p_0, q_0)$ and all $w \in A_{p/p_0} \cap RH_{(\frac{q_0}{p})'}$

$$\|g\|_{L^p(w)} \leq C(p, [w]_{A_{p/r}})\|f\|_{L^p(w)}.$$

Recall that $w \in A_{p/p_0} \cap RH_{(\frac{q_0}{p})'}$ can be reformulated as

$$w^{(q_0/p)'} \in A_{1+(\frac{p}{p_0}-1)(q_0/p)'}.$$

Remarks

There is a weighed Bochner-Riesz conjecture formulated as the following:

$$B^{\delta} : L^2(w) \to L^2(w), \qquad \text{for all } w^{\frac{n}{1+2\delta}} \in A_2.$$

Currently this is only proved for $w^{\frac{n}{1+2\delta}} \in A_2$ being radial (Duoandikoetxea-Moyua-Oruetxebarria-Seijo, 2008).

The basic ingredients of any of the proofs of the above extrapolation results are:

(1) Factorization.

(2) Construction of A_1 weights via the Rubio de Francia algorithm.

(3) (Sharp) bounds for the Hardy-Littlewood maximal function.

Point of departure

One of the weakness of Rubio de Francia's theory is that, it does not contain any information for the endpoint case. However, this is a matter of the A_p weights – we know that

$$\|(M \circ M)f\|_{L^p(w)} \lesssim \|f\|_{L^p(w)}$$

for all $1 < p < \infty$ with $w \in A_p$, but it is well-known that $M \circ M$ is not weak type $(1,1)$. Hence, to reach the endpoint we should need a larger class of weights.

Restricted A_p weights

We say w belongs to the restricted A_p class, which we denote by $A_p^{\mathcal{R}}$, if

$$[w]_{A_p^{\mathcal{R}}} = \sup_{E\subset Q} \frac{|E|}{|Q|}\Big(\frac{w(Q)}{w(E)}\Big)^{\frac{1}{p}}. \tag{a}$$

It is interesting to notice that $A_1^{\mathcal{R}} = A_1$. On the other hand, the definition can be equivalently defined as

$$\sup_Q \|\chi_Q w^{1/p}\|_{L^p(\frac{\mathrm{d}x}{|Q|})}\|\chi_Q w^{-1/p}\|_{L^{p',\infty}(\frac{\mathrm{d}x}{|Q|})} < \infty. \tag{b}$$

To see that (b) implies (a), just use Hölder's inequality

$$|E| = \|\chi_E\|_{L^{1,\infty}} \lesssim \|\chi_E w^{\frac{1}{p}}\|_{L^{p,\infty}}\|\chi_E w^{-\frac{1}{p}}\|_{L^{p',\infty}} \le w(E)^{\frac{1}{p}}\|\chi_Q w^{-\frac{1}{p}}\|_{L^{p',\infty}}$$

To see the converse, let $E_t := \{x \in Q : w^{-1/p} > t\}$. Then

$$\|\chi_Q w^{-1/p}\|_{L^{p',\infty}} = \sup_t t|E_t|^{1/p'} = \sup_t t\Big(\fint_{E_t} w\Big)^{1/p}\frac{|E_t|}{w(E_t)^{1/p}} \le \sup_t \frac{|E_t|}{w(E_t)^{1/p}}$$

Restricted A_p weights

By Kolmogorov's inequality, it is easy to see that $A_p \subset A_p^{\mathcal{R}} \subset A_{p+\varepsilon}$. An important result is the following

Theorem (Kerman-Torchinsky, 1982)

Let $1 \leq p < \infty$. *Then* $\|M\|_{L^{p,1}(w) \to L^{p,\infty}(w)} \sim [w]_{A_p^{\mathcal{R}}}$.

It would be ideally to formulate extrapolation theory for $A_p^{\mathcal{R}}$. However, currently the extrapolation is valid even with the smaller class $\widehat{A}_p$:

$$\widehat{A}_p := \{w = u(Mf)^{1-p} : u \in A_1,\ f \in L^1_{\mathrm{loc}}\}$$

It is still an open question whether $A_p^{\mathcal{R}} = \widehat{A}_p$ (for $p > 1$ since it is obvious for $p = 1$).

Extrapolation theorem for $\widehat{A}_p$ weights

Theorem (Carro-Grafakos-Soria, 2015)

Let T be an operator satisfying that, for some $p_0 > 1$,
$T : L^{p_0,1}(v) \to L^{p_0,\infty}(v)$ is bounded for every $v \in \widehat{A}_{p_0}$. Then
$T : L^{1,1/p_0}(v) \to L^{1,\infty}(v)$ is bounded for every $v \in A_1$.

Some remarks are given in below:

1. The operator T does not play any role
2. There is also a version which extrapolates from p_0 to some $p > 1$, for this purpose T is crucial to be sublinear (to apply interpolation).
3. In the proof, the construction of A_1 weight is not needed, so in particular the Rubio de Francia algorithm can be avoided.

Our motivation is to generalize this result to the multilinear setting.

Multilinear extrapolation

The first extrapolation theorem in the multilinear context is given by Grafakos and Martell (2004), for the weights $(w_1, \cdots, w_m)$ with $w_i \in A_{p_i}$. It states that, given an $m+1$ tuple of functions $(f_1, \cdots, f_m, f)$, if for some p_i, $1 \leq i \leq m$, with $1/p = \sum_{i=1}^{m} 1/p_i$

$$\|f\|_{L^p(w)} \lesssim \prod_{i=1}^{m} \|f_i\|_{L^{p_i}(w_i)}$$

holds for all $(w_1, \cdots, w_m)$ with $w_i \in A_{p_i}$, then for all q_i, $1 \leq i \leq m$, with $1/q = \sum_{i=1}^{m} 1/q_i$ and all $(w_1, \cdots, w_m)$ with $w_i \in A_{q_i}$,

$$\|f\|_{L^q(w)} \lesssim \prod_{i=1}^{m} \|f_i\|_{L^{q_i}(w_i)}.$$

A new proof of this result was given by Duoandikoetxea in 2010.

Multilinear extrapolation

The idea of Duoandikoetxea is the following: write

$$\|g\|_{L^p(w)} \lesssim \prod_{i=1}^m \|f_i\|_{L^{p_i}(w_i)}, \quad \text{where } w := \prod_{i=1}^m w_i^{p/p_i}$$

as

$$\|g \prod_{i=1}^{m-1} w_i^{1/p_i}\|_{L^p(w_m^{p/p_m})} \lesssim \Big(\prod_{i=1}^{m-1} \|f_i\|_{L^{p_i}(w_i)} \Big) \|f_m\|_{L^{p_m}(w_m)}.$$

Then since $w_m \in A_{p_m}$, it is easy to see that $w_m^{1/p_m} \in A_{p_m,p_m}$, here we say $v \in A_{p,r}$ if

$$\sup_Q \langle v^r \rangle_Q^{1/r} \langle v^{-p'} \rangle_Q^{1/p'} < \infty.$$

Using the off-diagonal extrapolation theorem one can extrapolate the inequality from $(p_1, \cdots, p_m) \to (p_1, \cdots, p_{m-1}, q_m)$. Repeating this process for other indices we will finally reach $(q_1, \cdots, q_m)$.

Off-diagonal extrapolation

Theorem (Duoandikoetxea, 2011)

Let μ be a doubling measure on $\mathbb{R}^n$, $n \geq 1$, and let (F, G) be a pair of non-negative Borel functions. Let $1 \leq p_0 < \infty$ and $0 < q_0, r_0 < \infty$ and assume that for all $w \in A_{p_0,r_0}(\mu)$ we have the inequality

$$\Big(\int_{\mathbb{R}^n} F^{q_0} w^{q_0} \mathrm{d}\mu \Big)^{\frac{1}{q_0}} \leq N([w]_{A_{p_0,r_0}}) \Big(\int_{\mathbb{R}^n} G^{p_0} w^{p_0} \mathrm{d}\mu \Big)^{\frac{1}{p_0}},$$

where N is an increasing function. Then there exists another increasing function $\widetilde{N}$ such that for all $1 < p < \infty$ and $0 < q, r < \infty$ verifying

$$\frac{1}{q} - \frac{1}{q_0} = \frac{1}{r} - \frac{1}{r_0} = \frac{1}{p} - \frac{1}{p_0},$$

for all $w \in A_{p,r}(\mu)$ we have

$$\Big(\int_{\mathbb{R}^n} F^{q} w^{q} \mathrm{d}\mu \Big)^{\frac{1}{q}} \leq \widetilde{N}([w]_{A_{p,r}}) \Big(\int_{\mathbb{R}^n} G^{p} w^{p} \mathrm{d}\mu \Big)^{\frac{1}{p}}.$$

Multiple $A_{\vec{p}}$ weights

In the linear case, A_p weights characterize the weighted boundedness of the Hardy-Littlewood maximal function, the Hilbert transform, and also the Riesz transforms. Namely,

$$w \in A_p \Leftrightarrow \|T\|_{L^p(w)\to L^p(w)} < \infty, \quad T \in \{M, H, R_j\}.$$

However, the class of weights $(A_{p_1}, \cdots, A_{p_m})$ does not characterize the weighted boundedness of the multilinear singular integrals such as the multilinear Riesz transforms.

Multiple $A_{\vec{p}}$ weights

The 'right thing' was found at 2009, by Lerner, Ombrosi, Pérez, Torres, and Trujillo-Gonzalez:

$$\vec{w} := (w_1, \cdots, w_m) \in A_{\vec{p}} \Leftrightarrow \|T\|_{L^{p_1}(w_1)\times\cdots\times L^{p_m}(w_m)\to L^p(w)} < \infty,$$

where T is either the multilinear maximal function, or the multilinear Riesz transforms. Here the multilinear maximal function is defined as

$$\mathcal{M}(f_1, \cdots, f_m) = \sup_{Q\ni x} \prod_{i=1}^{m} \fint_Q |f_i|$$

and we say $\vec{w} \in A_{\vec{p}}$ if

$$\sup_Q \Big(\fint_Q \prod_{i=1}^{m} w_i^{p/p_i} \Big) \prod_{i=1}^{m} \Big(\fint_Q w_i^{1-p_i'} \Big)^{p/p_i'} < \infty$$

It is also shown that the multilinear analogy of Coifman-Fefferman inequality holds with this new multilinear maximal function.

Multiple $A_{\vec{p}}$ weights

The multiple $A_{\vec{p}}$ weights can be more 'singular' than $(A_{p_1}, \cdots, A_{p_m})$ weights. For example, $(1, |x|^{-n}) \in A_{(1,1)}$ but the weight $|x|^{-n}$ is not even locally integrable.

The motivation to study the extrapolation theorem in the multilinear context is the following:

1. sparse domination gives estimate of the form $|\langle T(f_1, \cdots, f_n), g\rangle| \lesssim \cdots$
2. dyadic representation theorem gives $\langle T(f_1, \cdots, f_n), g\rangle = \cdots$

Both lead to only estimates in the Banach range. Hence it is crucial to use extrapolation to reach quasi-Banach range estimates.

Extrapolation for $A_{\vec{p}}$

Theorem (L.-Martell-Ombrosi, 2020)

Let $f, f_1, \cdots, f_m$ be an $m+1$ tuple of functions. Suppose that for some $p_1, \cdots, p_m$ with $1/p = \sum_i 1/p_i$,

$$\|f\|_{L^p(w)} \lesssim \prod_{i=1}^m \|f_i\|_{L^{p_i}(w_i)} \qquad \text{for every } \vec{w} \in A_{\vec{p}}.$$

Then for all $q_1, \cdots, q_m$ with $1/q = \sum_i 1/q_i$

$$\|f\|_{L^q(w)} \lesssim \prod_{i=1}^m \|f_i\|_{L^{q_i}(w_i)} \qquad \text{for every } \vec{w} \in A_{\vec{q}}.$$

The idea is to extend Duoandikoetxea's strategy to this setting. Namely, we fix the weights $w_1, \cdots, w_{m-1}$, and seek for a characterization of w_m such that $\vec{w} \in A_{\vec{p}}$.

There is also a multilinear Rubio de Francia algorithm constructed later by Zoe Nieraeth.

Our goal

The goal now is to establish a multilinear extension of Carro-Grafakos-Soria theorem, so that one can reach the endpoint. For this purpose, it is natural to define $A_{\vec{p}}^{\mathcal{R}}$ as

$$w \in A_{\vec{p}}^{\mathcal{R}} \Leftrightarrow \sup_{Q} \Big(\fint_{Q} \prod_{i=1}^{m} w_i^{p/p_i} \Big)^{1/p} \prod_{i=1}^{m} \| \chi_Q w_i^{-1/p_i} \|_{L^{p_i',\infty}(\frac{\mathrm{d}x}{|Q|})} < \infty.$$

It is indeed natural because the multilinear maximal function $\mathcal{M}$ is bounded from $L^{p_1,1}(w_1) \times \cdots \times L^{p_m,1}(w_m)$ to $L^{p,\infty}(\prod_{i=1}^m w_i^{p/p_i})$ if and only if $w \in A_{\vec{p}}^{\mathcal{R}}$.

It is unclear how to extend $\widehat{A}_p$ to the multilinear scenario.

Main obstacles

There are two basic obstacles:

1. The linear proof does not depend on Rubio de Francia algorithm, so this approach may not be appropriate for the multilinear case as well;
2. One would expect that Duoandikoetxea's idea works also here, however, in the strong norm case it is free to write

$$\|f\|_{L^p(\prod_{i=1}^m w_i^{p/p_i})} \quad \text{as} \quad \|f \prod_{i=1}^{m-1} w_i^{p/p_i}\|_{L^p(w_m^{p/p_m})}$$

However, for the weak type nom $\|f\|_{L^{p,\infty}(\prod_{i=1}^m w_i^{p/p_i})}$ this is not immediate.

First attempt

Nevertheless, the first result in this direction was given by Roure-Perdices recently

Theorem (Roure-Perdices, 2024)

Given measurable functions $f_1, \cdots, f_m$ and g. Suppose that for some exponents $\vec{p} = (p_1, 1, \cdots, 1)$ with $1 < p_1 < \infty$ and for all $\vec{w} = (w_1, \cdots, w_m) \in A^{\mathcal{R}}_{\vec{p}}$ we have

$$\|g\|_{L^{p,\infty}(\prod_{i=1}^m w_i^{p/p_i})} \lesssim \prod_{i=1}^m \|f_i\|_{L^{p_i,1}(w_i)}.$$

Then for all $(v_1, \cdots, v_m) \in A_{\frac{\vec{1}}{m}}$,

$$\|g\|_{L^{1/m,\infty}(\prod_{i=1}^m v_i^{1/m})} \lesssim \prod_{i=1}^m \|f_i\|_{L^{1,1/p_i}(v_i)}.$$

Main result

We extend Roure-Perdices' result to the full generality

Theorem (L.-Luque-Ombrosi, 2026+)

Given measurable functions $f_1, \cdots, f_m$ and g and $\alpha_i \in (0, p_i]$. Suppose that for some exponents $1 \leq p_1, \cdots, p_m < \infty$ with $1/p = \sum_i 1/p_i$ and for all $\vec{w} = (w_1, \cdots, w_m) \in A^{\mathcal{R}}_{\vec{p}}$ we have

$$\|g\|_{L^{p,\infty}(\prod_{i=1}^m w_i^{p/p_i})} \lesssim \prod_{i=1}^m \|f_i\|_{L^{p_i,\alpha_i}(w_i)}.$$

Then for all $(v_1, \cdots, v_m) \in A_{\frac{\vec{1}}{m}}$,

$$\|g\|_{L^{1/m,\infty}(\prod_{i=1}^m v_i^{1/m})} \lesssim \prod_{i=1}^m \|f_i\|_{L^{1,\alpha_i/p_i}(v_i)}.$$

Remark

The limited range version is also true, with more complicated statements and also the proof.

Ingredients of the proof

We follow Duoandikoetxea's strategy, and the novel idea is to use Sawyer type estimate.

Lemma (Cruz-Uribe, Martell, Pérez, 2005)

Let μ be a weight satisfying A_∞ condition. If $u \in A_1(\mu)$ and $uv \in A_\infty(\mu)$ then

$$\left\| \frac{M_\mu f}{v} \right\|_{L^{1,\infty}(uv\mu)} \lesssim \|f\|_{L^1(u\mu)}.$$

Ingredients of the proof

We also need the (semi)-characterization of $A_{\vec{p}}^{\mathcal{R}}$ so that we can apply the off-diagonal extrapolation theorem in this setting

Lemma (L.-Luque-Ombrosi, 2026+)

(a) *Let* $1 \leq p_i < \infty$ *for* $1 \leq i \leq m-1$ *and* $p_m = 1$ *and* $1/p = \sum_i 1/p_i$. *Then* $\vec{w} \in A_{\vec{p}}^{\mathcal{R}}$ *if and only if* $(w_1, \cdots, w_{m-1}, 1) \in A_{\vec{p}}^{\mathcal{R}}$ *and* $w_m^p \in A_1(\mu)$.

(b) *Let* $1 \leq p_1, \cdots, p_m < \infty$ *with* $1/p = \sum_i 1/p_i$. *Assume that* $(w_1, \cdots, w_{m-1}, 1) \in A^{\mathcal{R}}_{(p_1, \cdots, p_{m-1}, 1)}$ *and* $W^p \in \widehat{A}_{1+\frac{p}{p'_m}}(\mu)$. *If we set* $w_m = W^{p_m} \mu^{1-p_m}$, *then* $(w_1, \cdots, w_m) \in A_{\vec{p}}^{\mathcal{R}}$.

We remark that if $\vec{w} \in A_{\vec{p}}^{\mathcal{R}}$, then $w = \prod_{i=1}^m w_i^{p/p_i} \in A_\infty$. However, we do not know the information of w_i if $p_i \neq 1$.

Ingredients of the proof

Theorem (L.-Luque-Ombrosi, 2026+)

Let μ a weight belonging to the class A_∞. Let $1 \leq p_0 < \infty$ and $0 < q_0 < \infty$. Assume that for some family of nonnegative measurable functions f and g and for all $w \in \widehat{A}_{p_0,q_0}(\mu)$ we have

$$\|g\|_{L^{q_0,\infty}(w^{q_0}\mu)} \lesssim \|f\|_{L^{p_0,1}(w^{p_0}\mu^{1-p_0})}. \tag{12}$$

Then for all $1 < p < \infty$ and $0 < q < \infty$ such that $\frac{1}{q} - \frac{1}{q_0} = 1 - \frac{1}{p_0}$ holds, and all $w \in \hat{A}_{1,q}(\mu)$, we have

$$\|g\|_{L^{q,\infty}(w^q\mu)} \lesssim \|f\|_{L^{1,\frac{1}{p_0}}(w)}$$

Here we say that $v \in \widehat{A}_{r,q}(\mu)$ if and only if $v^q \in \widehat{A}_{1+\frac{q}{r'}}(\mu)$.

Applications

Consider

$$N(f,g)(x) = \sup_{\lambda>0} \lambda |\{y \in \mathbb{R}^n \setminus \{0\} : \frac{|f(x+y)||g(x+y)|}{|y|^{2n}} > \lambda\}|^2,$$

which is a bilinear variant of the maximal operators studied in ergodic theory.

$$N(f) = \sup_{\lambda>0} \lambda |\{y \in \mathbb{R}^n \setminus \{0\} : \frac{|f(x+y)|}{|y|^{n}} > \lambda\}|$$

It is well-known that N is not bounded from $L^1 \times L^1 \to L^{1/2,\infty}$ (since the linear N is not weak type (1,1)). However, by our extrapolation we could obtain

Theorem (L.-Luque-Ombrosi, 2026+)

Let $q > 1$ and $\vec{v} \in A_{\vec{\frac{1}{2}}}$. Then

$$\|N(f_1,f_2)\|_{L^{\frac{1}{2},\infty}(v_1^{1/2}v_2^{1/2})} \lesssim \prod_{i=1}^{2} \|f_i\|_{L^{1,\frac{1}{2q}}(v_i)},$$

Thank you!